Restricted Nonbranching Functional Exterior Matrix Product States for Continuous Interacting Fermions

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Abstract

Functional tensor networks expand each continuous particle coordinate in an orthonormal basis, contract the resulting operator matrices, and optimize a finite coefficient representation by automatic differentiation. Extending this construction to identical fermions is difficult: an ordinary particle-site tensor train carries exchange-statistics ranks, whereas generic compact exterior cores do not admit an assumed efficient exact contraction. We therefore study a restricted functional exterior matrix product state (FEMPS) with one globally conserved, nonbranching virtual path. The state is exactly a sum of K nonorthogonal Slater determinants in a continuous functional basis, embedded in the matrix-wedge definition. Norms, one-body and factorized two-body energies, and reverse-mode gradients are evaluated through batched K^2 determinant transitions, with inverse-based fast formulas and a singular-safe minor fallback. A conditioned generalized eigenproblem removes the linear determinant amplitudes during orbital optimization.

Numerical evidence is reported for harmonic and soft-Coulomb fermions. At $N = 4$, independent D and K axes, three-seed stability, and a $D = 8 \to 12$ continuation pass direct finite-basis configuration-interaction (CI), variance, norm, antisymmetry, time, and memory checks. At $N = 6, D = 10, K = 4$, three blind runs have CI errors 2.49×10^{-4}–4.03×10^{-4}, maximum variance 3.09×10^{-3}, and zero structural antisymmetry residual. Independent $N = 6$ axes further give monotone energies for $K = 1, 2, 4$ at $D = 10$ and $D = 8, 10, 12$ at $K = 4$; the final $D = 12$ CI error is 1.05×10^{-4}. A preregistered truth-free $D = 12$ adaptive continuation to $K = 6$ further reduces that error to 3.20×10^{-5} and the variance to 3.73×10^{-4}, while beating a same-budget cold $K = 6$ control. At fixed $D = 10, K = 4$ the ordinary particle-TT center rank is 24 for $N = 4$ and 80 for $N = 6$, while the FEMPS correlation control remains $K = 4$; matched kernel measurements expose the corresponding polynomial cost. The method is a restricted nonorthogonal multideterminant algorithm, not an efficient generic FEMPS contraction. Direct CI remains faster in the present small truth spaces, and no $N = 8$, asymptotic-scaling, DMRG-superiority, or categorical novelty claim is made.

1 Introduction

Hong et al. introduced a functional tensor-network route to continuous many-body Schrödinger equations [1]. For an orthonormal basis $\{\phi_p\}_{p=1}^{D}$, one writes

$$\Psi(x_1, \ldots, x_N) = \sum_{p_1,\ldots,p_N=1}^{D} C_{p_1\cdots p_N} \prod_{i=1}^{N} \phi_{p_i}(x_i). \tag{1}$$

Orthonormality converts continuum norms and local differential or multiplicative operators into finite tensor contractions. Hong et al. represented C by an MPS and optimized the Rayleigh quotient by automatic differentiation, leaving fermionic tensor networks as an open extension.

For identical fermions, C lies in $\Lambda^N V_D$ rather than a generic $V_D^{\otimes N}$. An ordinary particle-site TT can represent this alternating tensor, but its exact bonds include exchange multiplicities even for a single Slater determinant. Occupation-number MPS/DMRG, Hilbert-space MPS [2, 3], ordered first-quantized MPS [4], and an exterior particle-coordinate network are distinct factorizations and must retain their own names.

Exterior algebra enforces antisymmetry structurally, but compact exterior inputs do not by themselves imply tractable exact contraction. A companion structural analysis conditionally obstructs generic exact squared-norm contraction already at fixed small bond. Here that result is treated as an algorithm-design constraint: we impose a nonbranching global virtual label and test the resulting restricted class as a continuous variational solver.

The contribution is deliberately bounded:

1. an exact matrix-wedge embedding of a K-term nonorthogonal Slater sum;
2. polynomial value and reverse-mode determinant-transition contractions, including singular-safe validation;
3. a conditioned variable-projection optimizer with versioned checkpoints, residuals, and reproducibility records;
4. controlled $N = 2, 4, 6$ harmonic and soft-Coulomb benchmarks; and
5. a measured separation between the exterior exchange carrier and the correlation multiplicity K, together with its cost.

Finite sums of nonorthogonal Slaters and geminal states have substantial prior art [5, 6]. Accordingly, we do not claim a new determinant state class. The method-level content is the restricted exterior/functional integration, its contraction and AD implementation, and its controlled continuum-basis audit.

2 Functional exterior representation

2.1 Finite functional basis

Let $V_D = \mathrm{span}\{\phi_1, \dots, \phi_D\}$ with $\langle \phi_p, \phi_q \rangle = \delta_{pq}$. A normalized decomposable state is

$$\Phi[U] = u_1 \wedge \cdots \wedge u_N, \qquad U = (u_1, \dots, u_N) \in \mathbb{C}^{D\times N}, \quad U^\dagger U = I_N. \tag{2}$$

The normalized alternating embedding into labelled particle coordinates is

$$\mathcal{J}_N(u_1 \wedge \cdots \wedge u_N) = \frac{1}{\sqrt{N!}} \sum_{\pi\in S_N} \mathrm{sgn}(\pi) u_{\pi(1)} \otimes \cdots \otimes u_{\pi(N)}. \tag{3}$$

Antisymmetry is therefore exact before any numerical contraction.

2.2 Nonbranching diagonal-path FEMPS

A matrix-wedge FEMPS propagates virtual matrix indices while multiplying physical entries by exterior product. We restrict every core to be diagonal in one global path label $a = 1, \dots, K$. With boundary amplitudes c_a, the state becomes

$$\Psi_{K,D,N} = \sum_{a=1}^{K} c_a \, u_{a1} \wedge \cdots \wedge u_{aN} = \sum_{a=1}^{K} c_a \Phi_a. \tag{4}$$

No core can change a, hence there are K physical paths rather than K^{N-1} branching virtual histories. Equation (4) is an exact subclass of the original matrix-wedge definition and remains first quantized in the continuous coordinates of Eq. (1).

$K = 1$ is a single Slater. Increasing K enlarges a nonorthogonal secant span and can recover correlation. We call K the *correlation multiplicity* of this restricted parametrization; it is not a canonical entanglement spectrum. The stored nonlinear state has KDN complex orbital scalars before gauge removal.

3 Exact restricted contractions

Let $U_a \in \mathbb{C}^{D\times N}$ collect the orbitals of Φ_a and define the transition overlap

$$S_{ab} = U_a^\dagger U_b, \qquad \langle \Phi_a | \Phi_b \rangle = \det S_{ab}. \tag{5}$$

All norm and Hamiltonian matrices therefore have only K^2 transition pairs.

3.1 One-body transitions

For $\hat{H}_1 = \sum_i h(i)$, let $X_{ab} = U_a^\dagger h U_b$. If S_{ab} is well conditioned,

$$\langle \Phi_a | \hat{H}_1 | \Phi_b \rangle = \det(S_{ab}) \operatorname{Tr}(S_{ab}^{-1} X_{ab}). \tag{6}$$

Near singularity, the implementation replaces one column of S_{ab} at a time and sums the resulting determinants. This cofactor formula is valid even when S_{ab} is singular and is also an independent value/gradient reference.

3.2 Factorized two-body transitions

The admitted pair operator is written

$$\hat{V} = \frac{1}{2} \sum_{\ell=1}^{L} w_\ell \sum_{i<j} \left[L_\ell(i) R_\ell(j) + R_\ell(i) L_\ell(j) \right]. \tag{7}$$

For projected insertions $A_{ab}^\ell = U_a^\dagger L_\ell U_b$ and $B_{ab}^\ell = U_a^\dagger R_\ell U_b$, the well-conditioned mixed transition is

$$\det(S_{ab}) \left[\operatorname{Tr}(S_{ab}^{-1} A_{ab}^\ell) \operatorname{Tr}(S_{ab}^{-1} B_{ab}^\ell) - \operatorname{Tr}(S_{ab}^{-1} A_{ab}^\ell S_{ab}^{-1} B_{ab}^\ell) \right]. \tag{8}$$

The singular-safe route replaces two distinct columns and evaluates $K^2 LN(N-1)$ determinants. The auto path selects inverse formulas only after a conditioning gate. Well-conditioned K^2 pairs and all L factors are evaluated as batched solves; only singular pairs enter the column-replacement loop. Both branches remain differentiable under PyTorch AD, while the former pairwise implementation is retained as an independent reference.

3.3 Energy and complexity

With overlap and Hamiltonian matrices $S, H \in \mathbb{C}^{K\times K}$,

$$E(c, U) = \frac{c^\dagger H(U) c}{c^\dagger S(U) c}. \tag{9}$$

The stored orbitals require $O(KDN)$ memory. Dense one-body and factor actions cost $O(KD^2N)$ and $O(KLD^2N)$, followed by K^2 transition algebra. A conservative well-conditioned bound is polynomial in $K^2L(D^2N + N^3)$. The validation fallback performs $K^2LN(N-1)$ size-N determinants and is slower but still polynomial. Neither route enumerates all virtual paths or materializes D^N coefficients.

4 Variable-projection solver and scientific contract

Each U_a is kept in a Stiefel gauge by reduced QR, with the diagonal phase of R fixed deterministically. At every nonlinear orbital step, amplitudes are eliminated by the conditioned generalized Hermitian problem

$$H(U)c = E\, S(U)c. \tag{10}$$

Overlap directions below a relative threshold are discarded and the retained rank, raw/retained condition number, and generalized residual are reported. Adam with a cosine learning-rate schedule is followed by optional L-BFGS refinement. Reverse-mode derivatives propagate through QR, transitions, and the generalized eigenvalue.

The public result schema is version 2 and the checkpoint schema is version 1. Resume rejects a changed configuration, operator identity, or schema. Every call reports the structural antisymmetry residual, norm error, energy variance when exterior truth is enabled, elapsed time, sampled process RSS, conditioning, and exact operation counts. If D^N lies below a declared cap, a validation- only particle tensor is built and its adjacent-transposition residual is also reported. A null materialized residual means that the registered exponential validator was disabled; it never replaces the structural residual.

The machine-readable reproduction manifest records SHA-256 hashes, commands, seeds, tolerances, evidence labels, and independent verifiers. Claim identifiers used below refer to that manifest.

5 Models and reference calculations

We use harmonic-oscillator functions as the continuous basis. The first model is an analytically separable harmonically interacting fermion system, used for $N = 2$ validation and the initial $N = 4$ ladder. The nonquadratic model is

$$\hat{H} = \sum_{i=1}^{N} \left(-\frac{1}{2}\partial_{x_i}^2 + \frac{1}{2}x_i^2 \right) + \sum_{i<j} \frac{1}{\sqrt{(x_i - x_j)^2 + 1}}. \tag{11}$$

Soft-Coulomb integrals are projected by $Q = 128$ Gauss–Hermite quadrature. The production operator uses a physical $(p, r) \mid (q, s)$ SVD of the dense pair tensor; $D = 10$ and 12 retain ranks 19 and 23 with relative reconstruction errors 1.34×10^{-15} and 3.85×10^{-15}.

Direct references use the unfactorized four-index quadrature tensor and the Slater–Condon Hamiltonian in $\Lambda^N V_D$. CI is validation-only and never initializes a FEMPS run. At $N = 6, D = 10$ and 12 the exterior dimensions are $\binom{10}{6} = 210$ and $\binom{12}{6} = 924$, so exact diagonalization remains stronger than an approximate second-quantized DMRG control. DMRG is therefore deferred rather than relabeled as FEMPS.

6 Numerical evidence

All floating-point results in this section are numerical evidence. The full manifest contains eleven independently verified artifacts.

6.1 $N = 2$: analytic and finite-basis checks

The noninteracting $N = 2, D = 6, K = 1$ state gives $E = 2$ and zero variance exactly within complex128 arithmetic. For harmonic coupling $\kappa = 0.35$, increasing $K = 1, 2, 4$ at $D = 6$ lowers the energy, while $D = 4, 6, 8$ at $K = 4$ decreases the continuum-reference error. The final $D = 8, K = 4$ energy is 2.4557706074300585, with continuum-reference error 9.886×10^{-6}, same-basis error 9.797×10^{-6}, variance 8.423×10^{-5}, and zero antisymmetry residual (claim `n2_harmonic_ladder`).

The same registered ladder embeds the noninteracting $N = 4$ Slater at $D = 6, K = 1$, yielding $E = 8$ with zero variance and zero materialized antisymmetry residual. For the interacting harmonic model at $\kappa = 0.35$, three blind $N = 4, D = 6, K = 4$ runs agree within 1.782×10^{-12} and have maximum same-basis error 1.835×10^{-12} and maximum variance 2.272×10^{-11}. The $K = 1, 2, 4$ energy is nonincreasing at $D = 6$ and the continuum-reference error decreases over $D = 5, 6, 7$ at $K = 4$ (claim `n4_harmonic_e4_closure`).

Table 1: $N = 4, D = 6$ soft-Coulomb correlation axis.

K	Energy	Error versus CI	Variance
1	11.025279022118180	1.441×10^{-3}	6.360×10^{-3}
2	11.023866678350620	2.897×10^{-5}	2.009×10^{-4}
4	11.023837776253766	6.305×10^{-8}	3.147×10^{-7}

Evidence: `n4_soft_coulomb_transferability`.

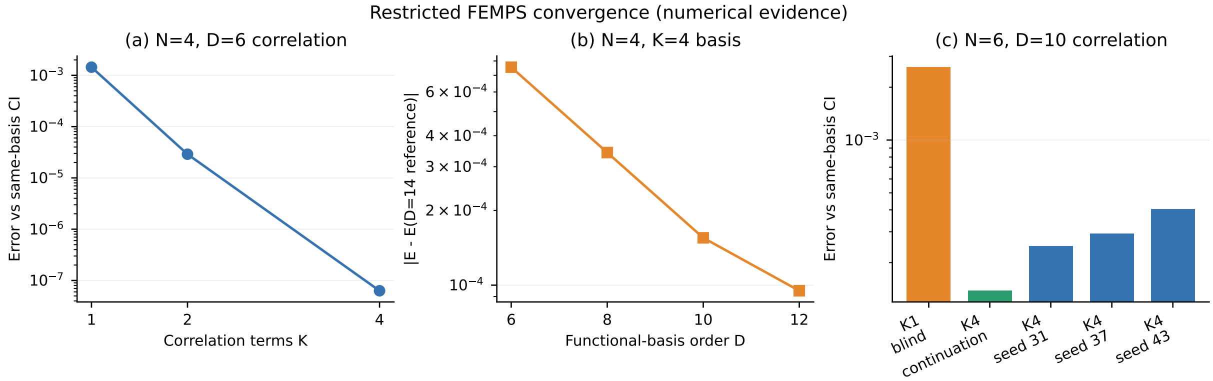

Figure 1: Restricted FEMPS convergence from manifest-hashed artifacts. Panel (a) is the N4 same-basis correlation error, panel (b) is the N4 functional-basis trend relative to a finite $D = 14$ numerical reference, and panel (c) separates N6 K1, K1-to-K4 continuation, and blind K4 starts. All values are numerical evidence.

Evidence: `n4_soft_coulomb_transferability`, `n4_soft_coulomb_basis_extension`, `n4_soft_coulomb_high_basis_correlation`, `n6_soft_coulomb_pilot`, and `n6_soft_coulomb_multiseed`.

6.2 $N = 4$: nonquadratic D and K control

For Eq. (11), Table 1 shows the independent $D = 6$ correlation axis. Increasing K reduces the same-basis CI error by more than four orders of magnitude.

At fixed $K = 4$, the absolute energy difference from a pre-existing $D = 14$ finite-basis numerical reference decreases from 7.549×10^{-4} at $D = 6$ to 9.494×10^{-5} at $D = 12$. This is not a continuum bound. Direct same-basis CI errors are 1.042×10^{-4} and 8.314×10^{-5} at $D = 10, 12$, with variances below 10^{-3} and zero structural/materialized antisymmetry residuals (claim `n4_soft_coulomb_basis_extension`).

Blind growth from $D = 12, K = 4$ to $K = 5$ lowers the same-basis CI error by 44.35% to 4.627×10^{-5} and the variance to 5.291×10^{-4}, while the measured time grows from 92.77 to 141.91 seconds (claim `n4_soft_coulomb_high_basis_correlation`). Three blind $D = 6$ starts and three truth-free $D = 6 \to 8$ continuations independently pass the registered stability gates.

6.3 $N = 6$: feasibility and blind stability

The resource-capped $N = 6, D = 10$ pilot first optimizes a blind $K = 1$ Slater and then preserves it while adding three seeded blind determinants. The CI energy is 25.049639839832263. The $K = 1$ error is 2.603×10^{-3}; $K = 4$ reduces it to 1.382×10^{-4} with variance 1.197×10^{-3}, zero materialized antisymmetry residual, 85.59-second runtime, and 798 MB sampled peak RSS (claim `n6_soft_coulomb_pilot`).

Three additional, genuinely blind K4 starts use identical 160-step Adam and 80-step L-BFGS budgets. Table 2 reports all runs rather than selecting the best initialization.

The energy spread is 1.537×10^{-4}; all four overlap directions are retained, norm and structural antisymmetry residuals are zero, and every point stays below 883 MB sampled peak RSS. Seed 31

Table 2: Blind $N = 6, D = 10, K = 4$ stability.

Seed	CI error	Variance	Condition number	Time (s)
31	2.494×10^{-4}	2.169×10^{-3}	3.394	112.14
37	2.932×10^{-4}	2.334×10^{-3}	3.209	105.06
43	4.031×10^{-4}	3.094×10^{-3}	4.424	105.92

Evidence: `n6_soft_coulomb_multiseed`.

Table 3: Independent $N = 6$ soft-Coulomb convergence axes.

Axis	Point	Energy	Error versus CI	Variance
K at $D = 10$	$K = 1$	25.052242782725	2.603×10^{-3}	1.478×10^{-2}
	$K = 2$	25.050276338618	6.365×10^{-4}	4.695×10^{-3}
	$K = 4$	25.049825287522	1.854×10^{-4}	1.749×10^{-3}
D at $K = 4$	$D = 8$	25.051264850894	3.196×10^{-5}	3.016×10^{-4}
	$D = 10$	25.049825287522	1.854×10^{-4}	1.749×10^{-3}
	$D = 12$	25.049471144618	1.047×10^{-4}	1.122×10^{-3}

Evidence: `n6_independent_dk_convergence`.

additionally materializes the one-million-coefficient tensor and gives zero antisymmetry residual.

Independent functional-basis and correlation axes are reported in Table 3. Every point uses 160 Adam and 80 L-BFGS steps. The $D = 12, K = 4$ point takes 107.66 seconds and 974,974,976 sampled RSS bytes. Its structural antisymmetry residual is zero; no 12^6 particle tensor is built.

Both the FEMPS and direct-CI absolute energies decrease over $D = 8, 10, 12$. The error relative to each same-basis CI result is not monotone because fixed $K = 4$ must approximate new correlation directions as the basis grows. This is an interpretable D/K tradeoff, not a continuum bound. At $D = 12, K = 4$ the state stores 292 complex FEMPS scalars; its ordinary particle-TT ranks are $(12, 60, 80, 60, 12)$ and require 132,768 scalars (claim `n6_independent_dk_convergence`).

ADR 0023 next freezes a truth-free adaptive $D = 12$ continuation. At each growth step, exactly 32 seeded Slater candidates are ranked only by the factorized determinant-transition generalized eigenvalue and balanced overlap conditioning. Dense CI is constructed only after the adaptive K5, adaptive K6, and cold K6 optimizations are frozen. Table 4 reports the result.

Adaptive K6 lowers the source energy by 7.271×10^{-5} and lies 2.441×10^{-4} below the same-budget cold K6. Its ordinary particle-TT ranks are $(12, 66, 120, 66, 12)$, requiring 209,376 scalars, while the restricted FEMPS state stores 438 orbital and amplitude scalars. This is a structural representation comparison, not a runtime advantage. K5, adaptive K6, and cold K6 take 6.72, 6.24, and 6.28 CPU seconds after vectorization; the 107.66 s source record predates Phase 33 and is not a matched timing comparator. The K6 pruning gate does not trigger (claim `n6_adaptive_correlation_growth`).

7 Exchange carrier, correlation multiplicity, and cost

At fixed $D = 10, K = 4, L = 19$, the exact structural counts grow as shown in Table 5. The $N = 6/N = 4$ ratio is 1.5 for stored orbitals and one-body determinants and 2.5 for two-body minors.

The singular-safe minor value/gradient timings grow by factors 2.115/2.232, close to the structural count after fixed overhead. The inverse-fast path grows by 1.239/1.024 over these two small points and accelerates the N6 minor path by 7.27/8.93. These are two-point CPU measurements, not asymptotic fits. Auto and minor transition matrices agree within 1.07×10^{-14}.

Table 4: Truth-free adaptive $N = 6, D = 12$ growth.

State	Energy	Error versus CI	Variance	Condition
Source $K = 4$	25.049471144618	1.047×10^{-4}	1.122×10^{-3}	2.111
Adaptive $K = 5$	25.049434533568	6.812×10^{-5}	7.747×10^{-4}	2.440
Adaptive $K = 6$	25.049398437461	3.202×10^{-5}	3.726×10^{-4}	2.760
Cold $K = 6$	25.049642546632	2.761×10^{-4}	2.659×10^{-3}	8.039

Evidence: `n6_adaptive_correlation_growth`.

Table 5: Matched production structure.

Quantity	$N = 4$	$N = 6$	Ratio
Stored orbital scalars	160	240	1.5
Transition pairs	16	16	1.0
One-body determinants	64	96	1.5
Two-body determinants	3,648	9,120	2.5
Enumerated virtual paths	0	0	–

Evidence: `matched_n4_n6_transition_cost`.

Phase 33 batches transition pairs and factor axes while retaining that pairwise implementation as a reference. Table 6 uses one fixed $N = 6, D = 10, K = 4, L = 19$ state and one matched blind optimization.

CPU batched/reference Hamiltonians differ by 8.90×10^{-16} and orbital gradients by 4.69×10^{-13}. CPU/Blackwell Hamiltonians differ by 1.43×10^{-14} and gradients by 3.68×10^{-12}. The matched optimizations give identical energy 25.050223374042; Blackwell finishes well inside 600 seconds with 20.7 MB peak device allocation. CPU remains faster by a factor 2.35 for the full registered solve, so Blackwell is an admitted backend but not an acceleration claim. Seed 3304 misses the separate Phase 29 CI-error quality control and is used only for backend parity; it does not replace the Phase 32 physics evidence.

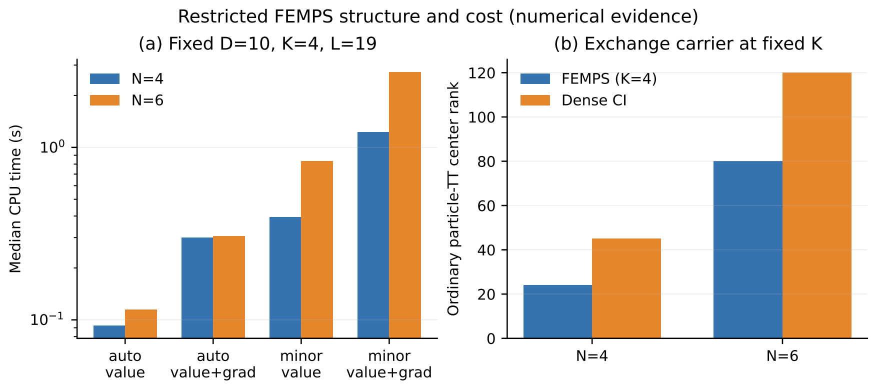

Figure 2: Matched N4/N6 kernel timings and ordinary particle-TT center ranks at fixed FEMPS $K = 4$. The exterior representation keeps the correlation multiplicity fixed while the labelled-particle exchange carrier grows. The plot does not imply runtime superiority over CI.

Evidence: `matched_n4_n6_transition_cost`.

Table 6: Vectorized transition and matched backend audit.

Path	FB kernel (s)	Reference speedup	160/80 solve (s)
CPU pairwise reference	0.28038	1.0	–
CPU batched	0.00803	34.9	5.08
Blackwell batched	0.01814	15.5	11.96

Evidence: `n6_vectorized_transition_backend`.

For $N = 4$ the ordinary particle-TT ranks are $(10, 24, 10)$ for the K4 state and $(10, 45, 10)$ for dense CI. At $N = 6$ they are $(10, 45, 80, 45, 10)$ and $(10, 45, 120, 45, 10)$. The FEMPS nonlinear correlation multiplicity remains $K = 4$ even though the ordinary center rank increases from 24 to 80. Exterior algebra does not remove exchange support; it prevents that support from being conflated with the adjustable correlation count.

8 Relation to other representations

The diagonal-path state is a nonorthogonal selected-CI-like Slater expansion, and determinant sums are established variational objects [5]. Finite AGP/AGP-CI methods likewise use polynomial pair-state transitions [6]. The present implementation differs in its explicit derivation as a restricted matrix-wedge FEMPS, direct use of the Hong et al. continuous functional-operator layer, and mandatory D/K/symmetry/cost audit. These differences do not establish categorical novelty of the underlying state family.

An ordinary particle TT [7] is an exact comparator in the labelled coordinate tensor product. Its ranks include the flat exchange spectrum of a Slater determinant, a standard fermionic reduced-density structure [8]. Occupation-space quantum-chemistry DMRG [2] and Hilbert-space MPS [3] operate in a different factorization. The ordered first-quantized route of Li–Waintal [4] removes labelled-particle redundancy by restricting to an ordered sector; it is an external first-quantized comparator, not a FEMPS backend.

9 Limitations and failure conditions

The strongest limitation is scope. Equation (4) forbids virtual branching and therefore does not solve generic matrix-wedge contraction. Expressivity is that of a K-Slater secant expansion; difficult states may require rapidly growing K. The measured cost is explicitly quadratic in K and linear in the two-body factor rank L before orbital and determinant costs.

Direct CI is faster in the current truth spaces, which extend to $\binom{12}{6} = 924$. The evidence demonstrates correctness, optimization stability, three-point N6 basis/correlation convergence, and a structural tradeoff, not an end-to-end speed or memory advantage. Adaptive K growth has one registered candidate-pool lineage and one cold control; it does not yet establish pool-seed stability or a universal stopping rule. Three basis points and two particle counts cannot establish asymptotic scaling; no $N = 8$ diagonal-path calculation is admitted. DMRG has been deferred because it would be weaker than exact CI in the present finite spaces; no FEMPS–DMRG performance inference follows.

The implementation uses exact deterministic contraction for the restricted class. A future truncation or stochastic estimator for generic cores would require a separate error, variance, failure-probability, and antisymmetry gate. No floating-point experiment in this paper is a theorem.

10 Conclusion

A globally conserved nonbranching path turns a matrix-wedge functional exterior network into a K-term nonorthogonal Slater sum. This restriction is strong enough to avoid known generic contraction obstructions while preserving first quantization, continuous functional bases, exact antisymmetry, and an adjustable multideterminant correlation span. Determinant transitions provide exact norms, one- and two-body energies, and AD gradients without virtual-path or particle-tensor enumeration in production.

Controlled N2/N4/N6 calculations establish a functioning restricted solver: energies improve systematically in D and K on registered axes, blind interacting runs are stable, and every accepted approximation/error/resource quantity is explicit. Truth-free fixed-pool growth at N6,D12 converts the correlation control into a measurable solver improvement, while its single-pool scope remains explicit. The ordinary particle-TT comparison makes the intended separation visible: exchange support grows while K remains the chosen correlation multiplicity. At the same time, current CI references are faster and the method is close to established nonorthogonal determinant expansions. The result is therefore a bounded algorithmic recovery of FEMPS, not a generic scalability claim.

A Reproduction and evidence mapping

The canonical short audit is

```
python scripts/build_phase30_reproduction_manifest.py
python scripts/verify_phase30_reproduction_manifest.py
python scripts/build_phase30_method_figures.py
python scripts/verify_phase30_method_figures.py
python scripts/verify_phase31_method_manuscript.py
python scripts/build_phase31_method_manuscript.py
python -m pytest -q
```

Long benchmark commands, seeds, tolerances, scientific boundaries, artifact hashes, and verifier modules are stored in `docs/experiments/results/phase30_reproduction_manifest.json`. Checkpoints are ignored regeneration artifacts and are not committed evidence.

Table 7: Manuscript numerical claims and manifest identifiers.

Identifier	Scope
`n2_harmonic_ladder`	N2 E1/E2 and noninteracting N4 E3
`n4_harmonic_e4_closure`	Interacting harmonic N4 closure
`n4_soft_coulomb_transferability`	N4 nonquadratic multiseed and D/K axes
`n4_soft_coulomb_basis_extension`	N4 D8-to-D12 continuation
`n4_soft_coulomb_high_basis_correlation`	N4 D12 K4-to-K5 growth
`n6_soft_coulomb_pilot`	N6 K1-to-K4 resource-capped pilot
`n6_soft_coulomb_multiseed`	N6 three-blind-seed stability
`matched_n4_n6_transition_cost`	Fixed-D/K/L N4-to-N6 cost audit
`n6_independent_dk_convergence`	N6 independent D8/D10/D12 and K1/K2/K4 axes
`n6_vectorized_transition_backend`	Batched CPU/Blackwell parity and matched solve
`n6_adaptive_correlation_growth`	Truth-free N6,D12 K4-to-K6 growth and cold K6 control

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